THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 13: **Statistics** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The class mark of the class 10–20 is: 1

(a) 10 **(b)** 15 **(c)** 20 **(d)** 30

2. The mode of grouped data is given by: 1

(a) $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$ **(b)** $l + \frac{f_1 - f_2}{2f_1 - f_0 - f_2} \times h$ **(c)** $l + \frac{f_1 - f_0}{f_0 + f_2} \times h$ **(d)** $l + \frac{f_1 + f_0}{2f_1 + f_0 + f_2} \times h$

3. If the mean is 20 and the median is 22, then by the empirical relation the mode is: 1

(a) 26 **(b)** 24 **(c)** 18 **(d)** 22

4. For the frequencies 0–10: 4, 10–20: 7, 20–30: 12, 30–40: 9, 40–50: 3, the modal class is: 1

(a) 10–20 **(b)** 20–30 **(c)** 30–40 **(d)** 40–50

5. A student states: “The median of grouped data is the mid-point of the middle class.” The correct response is: 1

(a) Correct **(b)** Incorrect — use $l + \frac{\frac{n}{2} - cf}{f} \times h$

(c) Correct for odd n **(d)** Cannot be determined

6. In the step-deviation method, $u = \frac{x - a}{h}$. If $a = 25$, $h = 10$ and $x = 45$, then u is: 1

(a) 2 **(b)** −2 **(c)** 20 **(d)** 0.2

7. For data with $n = 40$, Student A takes the median class as the class whose cumulative frequency first exceeds 20; Student B takes the class with the highest frequency. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. The mean of the first 10 natural numbers is: 1

(a) 5 **(b)** 5.5 **(c)** 6 **(d)** 10

9. If $\Sigma f_i = 25$ and $\Sigma f_i x_i = 1000$, the mean is: 1

(a) 40 **(b)** 25 **(c)** 1025 **(d)** 975

10. The most suitable measure to find the most popular shoe size sold in a shop is: 1

(a) mean **(b)** median **(c)** mode **(d)** range

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of the following data: 2

x	10	20	30	40
f	3	5	9	3

12. Find the mode of the data: 2

Runs	0–10	10–20	20–30	30–40	40–50	50–60
Batsmen	5	8	7	12	28	20

13. Find the median class and the modal class of: 2

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	3	9	15	30	18	5

14. If the mean of the following data is 15.5, find p: **2**

x	5	10	15	20	25
f	3	5	p	6	4

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the mean daily wage by the assumed mean method: **3**

Wages (₹)	100–120	120–140	140–160	160–180	180–200
Workers	12	14	8	6	10

16. Find the median of: **3**

Marks	0–10	10–20	20–30	30–40	40–50
Students	6	10	14	12	8

17. Find the mode of the ages of patients: **3**

Age (yrs)	5–15	15–25	25–35	35–45	45–55	55–65
Patients	6	11	21	23	14	5

OR

The median of the following distribution is 28.5 and $\Sigma f = 60$. Find x and y. **3**

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	5	x	20	15	y	5

18. A student found the mean of the following data to be 52. Verify his answer. **3**

Class	10–30	30–50	50–70	70–90	90–110	110–130
Frequency	5	8	12	20	3	2

OR

Find the mean literacy rate by the step-deviation method: **3**

Literacy (%)	45–55	55–65	65–75	75–85	85–95
Cities	3	10	11	8	3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. For the data below, find **5**

(i) the mean

(ii) the median

(iii) the mode using the empirical relationship.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	4	8	10	12	6

OR

The mean of the following distribution is 50 and $\Sigma f = 120$. Find the missing frequencies f_1 and f_2. **5**

Class	0–20	20–40	40–60	60–80	80–100
Frequency	17	f_1	32	f_2	19

20. **Case study:** A student says the mode of the data below is 42. **5**

(i) Find the actual mode.

(ii) Find the median.

(iii) Is the student correct?

Class	0–20	20–40	40–60	60–80	80–100	100–120	120–140
Frequency	6	8	10	12	6	5	3

OR

Find the median and mean height of the girls: **5**

Height (cm)	120–130	130–140	140–150	150–160	160–170
Girls	2	8	12	20	8

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND